

```
.color-transition {
```

```
background-color: lightblue;
```

```
transition: background-color 1s
```

```
}
```

```
.color-transition:hover {
```

```
background-color: white;
```

```
}
```



CSS

- CSS is the acronym for: 'Cascading Style Sheets'.
- CSS allows you to style your web pages.
- CSS describes how HTML elements should be displayed.

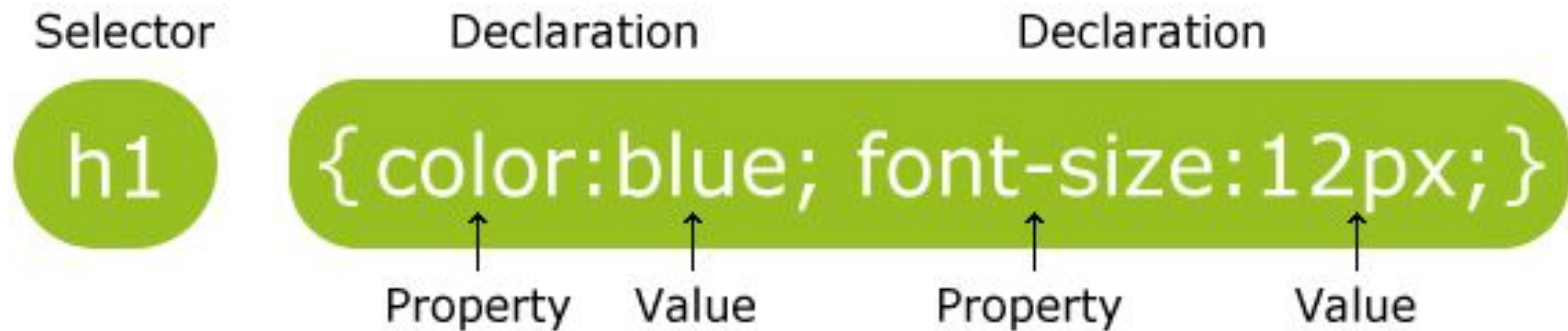
Examples

```
body {  
    background-color: grey;  
}
```

```
h1 {  
    color: white;  
    text-align: center;  
}
```

```
p {  
    font-family: verdana;  
    font-size: 25px;  
}
```

- CSS Syntax
- A CSS rule-set consists of a selector and a declaration block:



```
p {  
  color: red;  
  text-align: center;  
}
```

Inline Styles

- An inline style may be used to apply a unique style for a single element.

```
<h1 style="color:blue;margin-left:30px;">This is  
a heading</h1>
```

Internal Style Sheet

It is used if one single page has a unique style.

```
<head>
<style>
body {
  background-color: lightgrey;
}
h1 {
  color: maroon;
  margin-left: 50px;
}
</style>
</head>
```

Alternative code

```
<head>
```

```
<style>
```

```
@import "myStyleSheet1.css";
```

```
</style>
```

```
</head>
```

External Style Sheet

- With an external style sheet, you can change the look of an entire website by changing just one file

```
<head>
```

```
  <link rel="stylesheet" type="text/css"  
  href="mystyle.css">
```

```
</head
```


CSS Selectors

- Element Selector
- Id Selector
- Class Selector
- Universal Selector
- Group Selector
 - Universal selector *
 - Type selector elementname
 - Class selector .classname
 - ID selector #idname
 - Attribute selector [attr=value]

```
<style>  
input[type="text"]  
{  
  color:blue  
}  
</style>
```

CSS Element Selector

```
<style>
```

```
p{
```

```
  text-align: center;
```

```
  color: blue;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<p>This style will be applied on every paragraph</p>
```

id Selector

- The id selector uses the id attribute of an HTML element to select a specific element.
- The id of an element should be unique within a page so it is chosen to select a single, unique element.
- It is written with the hash character (#), followed by the id of the element.
- An id name cannot start with a number

```
<style>
```

```
#p1 {
```

```
    text-align: center;
```

```
    color: blue;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<p id="p1">Welcome to My Page</p>
```

```
<p>This paragraph will not be affected.</p>
```

```
</body>
```

```
</html>
```

```
<html>
<head>
<style>
#s1{
color:orange;
}
</style>
</head>
<body>
<p id=s1>
This is a paragraph
</p>
<h1>hello</h1>
</body>
</html>
```

class Selector

- To select elements with a specific class, write a period character, followed by the name of the class.

```
.center {  
    text-align: center;  
    color: red;  
}
```

```
<html>
<head>
<style>
.sty1
{
color:red;
text-align:center;
}
</style>
</head>
<body>
<p class=sty1>
This is a paragraph
</p>
<h1>hello</h1>
</body>
</html>
```


You can also specify that only specific HTML elements should be affected by a class.

```
p.center {  
    text-align: center;  
    color: red;  
}  
<p class="center">Hello</p>
```

HTML elements can also refer to more than one class.

- `<p class="center large">This paragraph refers to two classes.</p>`

Universal Selectors

- The * selector selects all elements.
- The * selector can also select all elements inside another element

```
* {  
color: red;  
}
```

- Select all elements inside <div> elements and set their background color to yellow:
- ```
div * {
 background-color: yellow;
}
```

```
<html>
<head>
<style>
div * {
 background-color: yellow;
}
</style>
</head>
<body>

<h1>Welcome to My Homepage</h1>

<div class="c1">
 <p>My page</p>
 <p>Welcome</p>
</div>

<p>End of My Page</p>

</body>
</html>
```

# Grouping Selectors

- It will be better to group the selectors, to minimize the code.
- To group selectors, separate each selector with a comma.

```
h1, h2, p {
 text-align: center;
 color: red;
}
```

# Descendant Selectors

- Used to apply a style rule to a particular element only when it lies inside a particular element.
- *element element*
- div p
- Selects all <p> elements inside <div> elements

```
<html>
<head>
<style>

```



# Note

```
div p {
 background-color: yellow;
}
```

- This code will also work if we write
- `<div>`
- `<h3><p>Hello</p></h3>`
- Because `<p>` is still inside `<div>` element

# Child Selectors

- Very similar to descendants but have different functionality.
- `body > p { color:grey; }`
- This rule will render all the paragraphs in black if they are direct child of `<body>` element. Other paragraphs put inside other elements like `<div>` or `<td>` would not have any effect of this rule.

```
<html>
<head>
<style>
div> p {
 background-color: Orange;
}
</style>
</head>
<body>
<h1>Welcome to My Homepage</h1>
<div>
 <h2><p>its my training period</p></h2>
 <p> Saket Nagar ,Kanpur</p>
</div>
<p>Course PGDM | MCA</p>
</body>
</html>
```



## Clear Comparison Table

Feature	Descendant Selector ( <input type="checkbox"/> )	Child Selector ( <input checked="" type="checkbox"/> )
Symbol	Space	<input checked="" type="checkbox"/>
Selects	All levels inside parent	Only direct children
Strictness	Less strict	More strict
Example	<code>div p</code>	<code>div &gt; p</code>

# [*attribute*] Selector

- The [*attribute*] selector is used to select elements with the specified attribute.
- Syntax
- Element [*attribute*] {  
    *css declarations*;  
}

example-

```
a[href^="https"] {
 color: green;
}
```

Selects all **<a>** links whose **href** starts with **https**

```
<html>
<head>
<style>
input[type = "text"]
{
 color: red;
}
</style>
</head>
<body>
<form>
<input type="text" name="t1">
</form>
</body>
</html>
```

# CSS Pseudo-Elements

Pseudo-elements let you style **specific parts of an element** or **insert generated content** without modifying HTML.

They use the double-colon syntax: `::`.

- `::after`
- `::before`
- `::first-letter`

Note -Requires `content` property



Inserts generated content **after**, **before** an element's content.

```
p::after {
 content: " - Important";
}

p::before {
 content: url('images/image.png');
}
```

**::selection**

Styles the portion of text selected by the user.

```
::selection {
 background: yellow;
 color: black;
}
```

Works when user drags mouse to select text.

```
<html>
<head>
<style>
p::first-letter {
 font-size: 200%;
 color: red;
}

p::before {
 content: "Note-";
}
</style>
</head>
<body>
<p>it's the example window.</p>
<p></p>
</body>
</html>
```

Pseudo-Element	Purpose
<code>::before</code>	Insert content before element
<code>::after</code>	Insert content after element
<code>::first-letter</code>	Style first letter
<code>::first-line</code>	Style first line
<code>::selection</code>	Style selected text
<code>::marker</code>	Style list bullet/number
<code>::placeholder</code>	Style input placeholder
<code>::file-selector-button</code>	Style file input button
<code>::backdrop</code>	Style modal background
<code>::cue</code>	Style video captions
<code>::slotted()</code>	Style slotted content

## `::marker`

Styles list item markers (bullets/numbers).

```
li::marker {
 color: red;
 font-size: 20px;}
```

Works only on:

```

<summary>
```

## `::placeholder`

Styles placeholder text inside input/textarea.

```
input::placeholder {
 color: gray;
 font-style: italic;}
```

## `::file-selector-button`

Styles the button inside file input.

```
input[type="file"]::file-selector-button {
 background: blue;
 color: white;}
```

## `::backdrop`

Styles the background behind elements in fullscreen or dialog mode.

```
dialog::backdrop {
 background: rgba(0,0,0,0.5);
}Used with:<dialog>
```

# Pseudo-Class, When It Applies

## Example

Pseudo-Class	When It Applies	Example
<code>:hover</code>	Mouse over element	<code>button:hover</code>
<code>:active</code>	While clicking element	<code>button:active</code>
<code>:focus</code>	Element is focused	<code>input:focus</code>
<code>:focus-visible</code>	Focus via keyboard navigation	<code>button:focus-visible</code>
<code>:visited</code>	Visited links	<code>a:visited</code>
<code>:link</code>	Unvisited links	<code>a:link</code>

```
button:hover {
 background: black;
 color: white;
}
```

```
input:focus {
 outline: none;
 border-color: blue;
}
```

Pseudo-Class	Selects	Example
<code>:first-child</code>	First child of parent	<code>p:first-child</code>
<code>:last-child</code>	Last child of parent	<code>p:last-child</code>
<code>:only-child</code>	Only child of parent	<code>div:only-child</code>
<code>:nth-child(n)</code>	Specific child number	<code>li:nth-child(2)</code>
<code>:nth-last-child(n)</code>	From bottom	<code>li:nth-last-child(1)</code>
<code>:first-of-type</code>	First of its type	<code>p:first-of-type</code>
<code>:last-of-type</code>	Last of its type	<code>p:last-of-type</code>
<code>:only-of-type</code>	Only element of its type	<code>p:only-of-type</code>
<code>:nth-of-type(n)</code>	Specific type count	<code>p:nth-of-type(2)</code>
<code>:nth-last-of-type(n)</code>	From bottom (type based)	<code>p:nth-last-of-type(1)</code>

# CSS Comments

```
/* This is a single-line comment */
```

```
/* This is
 a multi-line
 comment */
```

# Ways to Insert CSS in HTML

1. Inline style
2. Internal style sheet
3. External style sheet

# CSS Rules Overriding

- Any inline style sheet takes highest priority. So, it will override any rule defined in `<style>...</style>` tags or rules defined in any external style sheet file.
- Any rule defined in `<style>...</style>` tags will override rules defined in any external style sheet file.
- Any rule defined in external style sheet file takes lowest priority, and rules defined in this file will be applied only when above two rules are not applicable.



# CSS Backgrounds

CSS background properties:

- background-color
- background-image
- background-repeat
- background-attachment
- background-position

Property	Purpose	Common Values
background-color	Sets background color	red, #000, rgb()
background-image	Adds background image	url("image.jpg")
background-repeat	Controls image repetition	repeat, no-repeat, repeat-x, repeat-y
background-attachment	Scroll behavior of image	scroll, fixed, local
background-position	Sets image position	left, center, top, 50% 50%

## HTML

```
<div class="box">
 Background Example
</div>
```

## CSS

```
.box {
 height: 300px;

 background-color: lightblue; /* Background color */
 background-image: url("background.jpg"); /* Image */
 background-repeat: no-repeat; /* No repeating */
 background-attachment: fixed; /* Fixed while scrolling */
 background-position: center; /* Centered image */
}
```

## WITH SHORTHAND PROPERTY

**background:** lightblue url("background.jpg") no-repeat fixed center;

## SYNTAX

Color → image → repeat → attachment → position

# background-position

- The position of the image is specified by the background-position property:
- background-position: left top
- background-position: bottom right

# CSS Backgrounds

```
body {
 background-color: green;
 background-image: url("paper.gif");
 background-repeat: repeat-x | repeat-y |
no-repeat | repeat (default);
 background-position: left top;
 background-attachment: fixed /scroll;
}
```

# Repeat Background Image

- repeat-x will repeat a background image only horizontally.
- background-repeat: repeat-x;
- Repeat-y will repeat a background image only vertically.
- background-repeat: repeat-y;
- background-repeat: repeat; {default}
- background-repeat: no-repeat;

# Background position

- background-position: right top;
- background-position: bottom right;
- background-position: 100px;
- background-position: 50px 70px;

# CSS BORDER

- The *border* properties allow you to specify how the border of the box representing an element should look.
- There are three properties of a border you can change:
- The **border-color** specifies the color of a border.
- The **border-style** specifies whether a border should be solid, dashed line, double line, etc.
- The **border-width** specifies the width of a border.
- The **border-radius** property defines the radius of the element's corners.

# Properties

Property	Purpose	Common Values	Example Value
<code>border-color</code>	Sets the color of the border	<code>red, #000, rgb()</code>	<code>blue</code>
<code>border-style</code>	Defines the border pattern	<code>solid, dashed, dotted, double, groove, ridge, inset, outset, none</code>	<code>solid</code>
<code>border-width</code>	Sets thickness of border	<code>px, thin, medium, thick</code>	<code>3px</code>
<code>border-radius</code>	Rounds the corners	<code>px, %</code>	<code>10px</code>



# border-style Property

- **none:** No border. (Equivalent of border-width:0;)
- **solid:** Border is a single solid line.
- **dotted:** Border is a series of dots.
- **dashed:** Border is a series of short lines.
- **double:** Border is two solid lines.
- **groove:** Border looks as though it is carved into the page.
- **ridge:** Border looks the opposite of groove.
- **inset:** Border makes the box look like it is embedded in the page.
- **outset:** Border makes the box look like it is coming out of the canvas.
- **hidden:** Same as none, except in terms of border-conflict resolution for table elements.

# Example

```
p {
 border-style: solid;
 border-width: 5px;
 border-color: green;
}
```

# Example

```
<style>
```

```
div {
```

```
 border: 1px solid black;
```

```
 background-color: grey;
```

```
 padding-top: 50px;
```

```
 padding-right: 30px;
```

```
 padding-bottom: 50px;
```

```
 padding-left: 80px;
```

```
}
```

```
</style>
```

# Border - Shorthand Property

The **border** property is a **shorthand property** in CSS. It combines **three properties** into one single line:

```
border: border-width border-style border-color;
```

```
<style>
```

```
p{
```

```
border: 2px solid blue;
```

```
}
```

```
</style>
```

# CSS - Fonts

- The **font-family** property is used to change the face of a font.
- The **font-style** property is used to make a font italic or oblique.
- The **font-variant** property is used to create a small-caps effect.
- The **font-weight** property is used to increase or decrease how bold or light a font appears.
- The **font-size** property is used to increase or decrease the size of a font.

- Font-size :  
xx-small/small/medium/large/x-large/xx-large  
/no. in px or %
- `<p style="font-size:x-large;">`
- `<p style="font-size:200%;">`
- `<p style="font-size:20px;">`

- Font Style : italic /oblique/normal;
- CSS Font Variant :small-caps/normal;
- Font Weight :normal, bold, bolder, lighter or value (100-900);

```
p {
 font-family: verdana;
 font-size: 20px;
 font-style:italic;
 font-variant:small-caps;
 font-weight:bold | 500;
}
```



# CSS Text Properties

- Color
- Color:red;
- text-align: left/right/center/justify;
- text-decoration: overline/underline/line-through;
- text-transform: uppercase/lowercase/capitalize;
- letter-spacing: 3px;
- line-height: 0.8;
- text-indent: 70px;

- The text-indent property is used to specify the indentation of the first line of a text:
- text-indent: 50px;
- text-shadow: 3px(Hori Shadow) 2px (Vert Shadow) red;
- direction: rtl/ltr;
- word-spacing: 10px | -10px;

# CSS TEXT FORMATTING

```
p {
color: blue;
text-align: center/left/right/justify;
text-decoration: overline/underline/line-through;
text-transform: uppercase/lowercase/capitalize;
text-indent: 50px;
letter-spacing: 3px;
line-height: 1.0;
direction: rtl;
word-spacing: 10px ;
text-shadow: 3px 2px red;
}
```

## CSS Margins

- The CSS margin properties are used to generate space around elements.
- Units : px,pt,cm,%
- Margin-This property is used to set all the properties in one declaration.
- `margin-left /margin-right/margin-top/margin-bottom`

CSS has properties for specifying the margin for each side of an element:

- [margin-top](#) - sets the top margin of an element
- [margin-right](#) - sets the right margin of an element
- [margin-bottom](#) - sets the bottom margin of an element

- [margin-left](#) - sets the left margin of an element

- `auto` - the browser calculates the margin
- *length* - specifies a margin in px, pt, cm, etc.
- `%` - specifies a margin in % of the width of the containing element
- `inherit` - specifies that the margin should be inherited from the parent element

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
div {
 border: 1px solid black;
 margin-top: 100px;
 margin-bottom: 100px;
 margin-right: 150px;
 margin-left: 80px;
 background-color: lightblue;
}
</style>
</head>
<body>
<h2>Using individual margin properties</h2>
<div>This div element has a top margin of
100px, a right margin of 150px, a bottom
margin of 100px, and a left margin of
80px.</div>

</body>
```

```
</html>
```

Shorthand Property

[margin-top](#) [margin-right](#) [margin-bottom](#)  
[margin-left](#)

**margin: 25px 50px 75px 100px;**

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
div {
```

```
 border: 1px solid black;
```

```
 margin: 25px 50px 75px 100px;
```

```
 background-color: lightblue;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

<h2>The margin shorthand property - 4 values</h2>

<div>This div element has a top margin of 25px, a right margin of 50px, a bottom margin of 75px, and a left margin of 100px.</div>

<hr>

</body>

</html>

**Negative values are allowed. Default value:0**

**All the margin properties can have the following values:**

- **auto** - the browser calculates the margin and horizontally center the element and the remaining space will be split equally between the left and right margins

• **length** - specifies a margin in px, pt, cm, etc.

If the [margin](#) property has one value:

- **margin: 25px;**
  - all four margins are 25px

**margin: 25px 50px;**

- top and bottom margins are 25px
- right and left margins are 50px

The inherit Value

You can set the [margin](#) property to inherit to let the margin be inherited from the parent element.

<!DOCTYPE html>

<html>

<head>

```
<style>
```

```
div {
```

```
 border: 1px solid red;
```

```
 margin-left: 100px;
```

```
}
```

```
p.ex1 {
```

```
 margin-left: inherit;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2>Use of the inherit value</h2>
```

```
<p>Let the left margin be inherited
from the parent element:</p>
```

```
<div>
```

```
<p class="ex1">This paragraph has an
inherited left margin (from the div
element).</p>
```

```
</div>
```

```
</body>
```

```
</html>
```